Cryptography and Cybersecurity

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Homework 1:

Implementation Miller-Rabin Primality Test

Implementation Miller-Rabin Primality Test

Using the Matlab environment or Python language (student's choice), write a code to implement Miller-Rabin's primality test.
The code must:

1. take as input an integer number.
2. determine whether the number is composite or whether it may be prime.
3. offer the possibility to repeat the test with a different base of the test a.

After implementing the code, check that:

1. five randomly selected prime numbers pass the test.
2. five randomly selected composite numbers do not pass the test.
3. the composite number 221=13*17 passes the test if the base a is chosen as 1, 21

Miller-Rabin's Test

TEST(n)

1. Find integers k, q with $k>0$ and q odd such that $(n-1=2^k q)$
2. Choose a random integer a, such that $1<=a<=n-1$
3. **if** $a^q \bmod n = 1$ **then** return 'maybe prime'
4. **for** $j=0$ **to** $k-1$ **do**
5. **if** $a^{q2^j} \bmod n = n-1$ **then** return 'maybe prime'
6. return 'composite'

Useful MATLAB Commands

1. **primes**: generates prime numbers
2. **logical**: defines the logical type
3. **input**: takes an input from the keyboard
4. **disp**: display an array
5. **de2bi**: converts a decimal value into binary
6. **bi2de**: converts a binary number into a decimal value
7. **double**: converts a string of ASCII characters into an array of integer numbers
8. **powermod**: modular exponentiation
9. **char**: converts an array of integer numbers into the corresponding ASCII character
10. **xor**: logical OR operation
11. **circshift**: circular shift of the entries of a vector
12. **load**: load a MATLAB file
13. **save**: saves a MATLAB file

If you are using Python similar commands can be found in the numpy library!